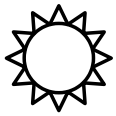
PHYSICAL AND CHEMICAL SCIENCE

Light and Sound

Light sources, darkness, shadows, and sounds made by vibrations


Sun


torch


candle


drum


bell

In this pack

Learn it: Light, dark and sound ★☆☆
Fact check: Light and sound facts ★☆☆
Think it through: Explain it ★★☆
Investigate: Shadow size ★★☆
Real-world case study: Shadow puppets ★★★
Review and challenge: Show what you know ★★★

This pack belongs to: _____ **Class:** _____

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LEARN IT · C3 · LIGHT AND SOUND

Light, dark and sound

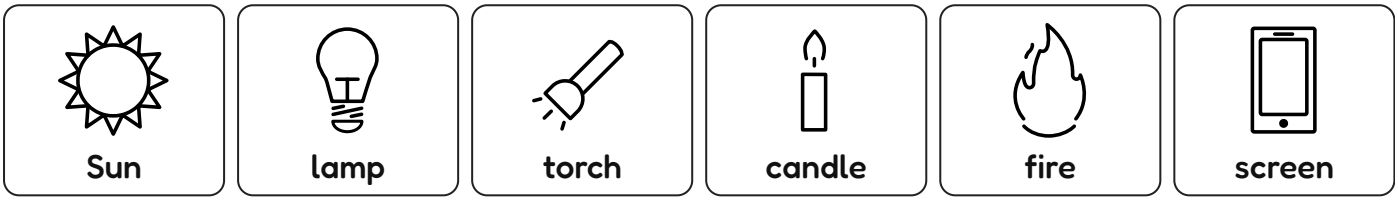
★☆☆ Grade 1

Name: Class: Date:

Learning goals

- Name light sources and explain darkness.
- Explain shadows and how sounds are made.

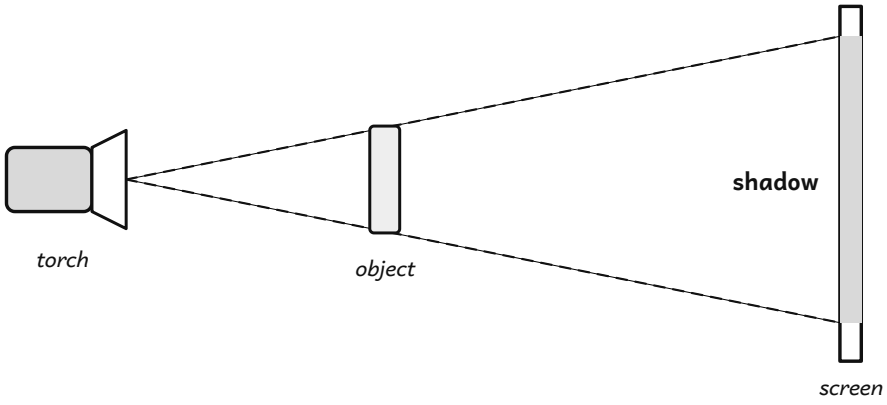
We need **light** to see. Things that give out light are **light sources**. The biggest is the **Sun**.



Darkness is when there is no light. In a dark room we cannot see anything. The Moon is not a light source: it reflects light from the Sun.

Shadows

When something blocks the light, a dark **shadow** forms behind it. Shadows have the same shape as the thing blocking the light.



An object blocks the light and makes a shadow.

Sound

Sounds are made when things **vibrate**: shake very fast. A drum skin vibrates when you hit it. A string vibrates when you pluck it. We hear sounds with our **ears**.



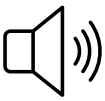
drum



guitar



bell



speaker



barking dog

Key words

light source	something that gives out light
shadow	a dark shape where light is blocked
vibrate	shake very fast

FACT CHECK · C3 · LIGHT AND SOUND

Light and sound facts

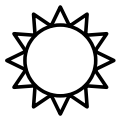
★☆☆ Grade 1

Name: Class: Date:

Learning goals

- Find light sources.
- Sort sources.

1 Circle the light sources.


Sun


Moon


candle


mirror


torch


bottle

2 True or false?

Statement	True	False	If false, correct it
1. The Moon is a light source.			
2. A shadow forms when light is blocked.			
3. Sounds are made by vibrations.			

Explain it

★★☆
Grade 1

Name: Class: Date:

Learning goals

- Explain shadows and sounds.

1 Why can't we see in a dark room?

2 Why does a rubber band make a sound when you pluck it?

3 Draw a shadow puppet and its shadow.

Shadow size

★★☆ Grade 1

Name: Class: Date:

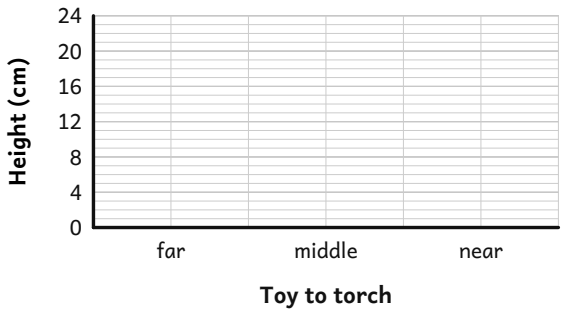
Learning goals

- Change a shadow.
- Read a bar chart.

Class 1 moved a toy closer to a torch and measured the size of its shadow on the wall.

1 Our results. Draw the bars.

Toy to torch	far	middle	near
Shadow height (cm)	8	14	25



2 What happens to the shadow when the toy is near the torch?

REAL-WORLD CASE STUDY · C3 · LIGHT AND SOUND

Shadow puppets

★★★
Grade 1

Name: Class: Date:

Learning goals

- Read about shadow puppets.

At the village fair, a man put on a **shadow puppet** show. He held cut-out puppets between a lamp and a white cloth.

The audience saw dark shadows of a tiger, a king and a boat on the cloth. A drummer played music. Everyone cheered!

1 What was the light source?

2 Why were the shadows on the cloth?

3 How did the drummer make sound?

REVIEW AND CHALLENGE · C3 · LIGHT AND SOUND

Show what you know

★★★
Grade 1

Name: Class: Date:

Learning goals

- Use light and sound words.

1 Find the words.

S	O	I	W	E	G	I
D	N	P	H	F	O	H
C	R	S	P	K	L	K
E	H	U	F	J	I	E
G	A	N	M	I	G	F
A	U	R	I	E	H	A
D	A	R	K	O	T	M

dark drum ear light sun

2 A shadow forms when...

○ light is blocked ○ it is very loud

How am I doing? Colour or tick a circle.

I can...	Not yet	Nearly	Got it!
name light sources	○	○	○
explain how a shadow forms	○	○	○
say how sounds are made	○	○	○